EMS-Whisper: A Speech Recognition Module for the Emergency Medical Services

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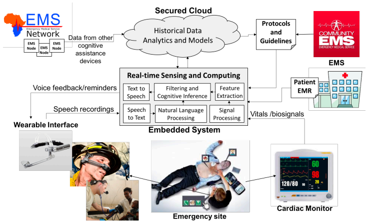
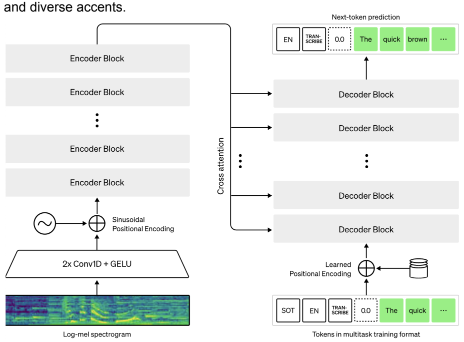
Overview

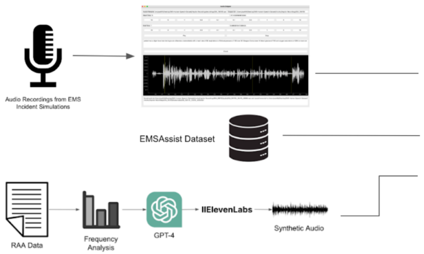


The EMS-Whisper project aims to create a real-time speech recognition module for transcribing audio in emergency situations. This module will be part of the larger CognitiveEMS system, an AI assistant designed to assist first responders by providing reminders and feedback during emergencies. It can also generate transcripts for easier post-incident reporting. For this use case, the speech recognition module needs to be accurate and robust in noisy environments typical of emergency scenes.

EMS-Whisper builds upon OpenAI's Whisper speech recognition model[1], which adopts the encoder-decoder Transformer[2] architecture. The Whisper models, ranging in size from 39M parameters to 1.55B parameters, were trained on 680k hours of internet audio, making them resilient to noisy audio and diverse accents.

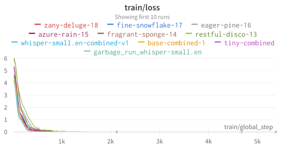


Data and Fine-tuning

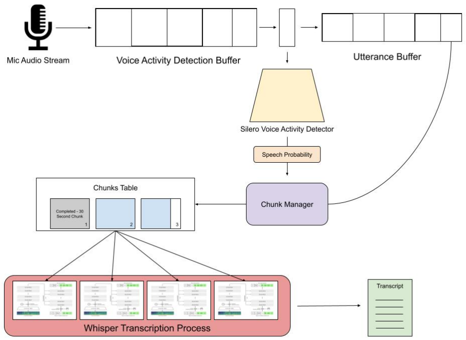


Dataset Name	Train	Validation	Testing
UVA-Chunked	567	121	123
EMSAssist	898	224	600
Synthetic	405	86	88
Total	**1870**	**431**	**811**

We discovered that the Whisper models exhibited reasonable performance on evaluation datasets, but made crucial errors when transcribing EMS terminology. To improve performance within this specific domain, we finetuned the models using 3 subdatasets: 1) **UVA-Chunked**: We manually segmented audio snippets from recordings of EMS responders simulating emergency scenarios. 2) **EMSAssist**[3]**:** We incorporated a public dataset that contained medical terminology from another research group. 3) **Synthetic Dataset**: We generated synthetic audio using ElevenLab's audio generation API and paired it with synthetic text from OpenAI's GPT-4, prompted by procedures, medications, call types, and impressions extracted from 35,000 ePCR reports obtained from a local EMS agency.



Real-time Processing



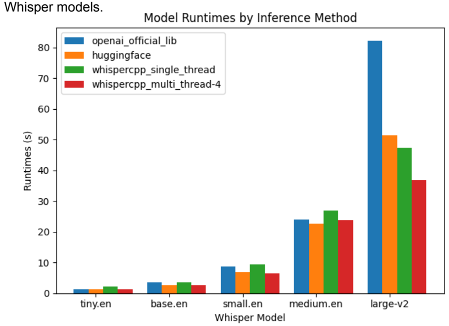
Once we had fine-tuned the model weights, we designed a real-time processing module that integrates with the models to provide continuous transcription results within the CognitiveEMS pipeline. This module functions by continuously receiving audio from a microphone stream and processing audio frames using the Silero open-source deep learning voice activity detection (VAD) model[5]. Depending on whether speech is detected, we add the frame to audio chunk buffers that the Whisper models ingest to provide ongoing transcription results.

Since Whisper follows a Seq2Seq Transformer architecture, it can effectively utilize contextual information within a single chunk of audio, which can extend up to 30 seconds. We tailored our real-time module to take full advantage of this design, allowing EMS responders to adjust the maximum chunk length as needed. All these components operate concurrently within a multi-process architecture, enabling independent rates of audio frame ingestion and transcription. Importantly, this real-time module remains adaptable to different runtime methods and Whisper models, so EMS workers can customize it to suit their computational constraints.

Evaluation

WER Table	UVA-Chunked	Synthetic-EMS	EMSAssist
tiny.en	0.188	0.048	0.280
tiny.en-finetuned	0.148	0.032	0.106
base.en	0.158	0.038	0.184
base.en-finetuned	0.103	0.023	**0.060**
small.en	0.119	0.030	0.128
small.en-finetuned	**0.079**	**0.014**	0.064
medium.en	0.090	0.024	0.102
large-v2	0.093	0.021	0.076
EMSConformer	N/A	N/A	0.077

The fine-tuned Whisper models achieved lower Word Error Rates (WER) than their original counterparts. Furthermore, the fine-tuned versions of the small.en and base.en models outperform the previous state-of-the-art model (large-v2), while requiring ~6x and ~16x less compute. (Note that WER results were calculated after token normalization). We also measured compute time, CPU usage, and memory usage for various combinations of models and runtime methods. The best runtime performances were obtained with HuggingFace's[4] open-source Whisper implementation and whisper.cpp's[6] multi-core implementation. We offer users the option to choose either of these methods for real-time inference with any of the Whisper models.



Citations: Radford, Alec, et al. "Robust speech recognition via large-scale weak supervision." *International Conference on Machine Learning*. PMLR, 2023. 2. Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017). 3. Jin, Liuyi et al. "EMSAssist: An End-to-End Mobile Voice Assistant at the Edge for Emergency Medical Services." *Proceedings of the 21st Annual International Conference on Mobile Systems, Applications and Services*. Association for Computing Machinery, 2023. 4. Wolf, Thomas, et al. "Huggingface's transformers: State-of-the-art natural language processing." *arXiv preprint arXiv:1910.03771* (2019). 5. Silero Team "Silero VAD: pre-trained enterprise-grade Voice Activity Detector (VAD), Number Detector and Language Classifier," https://github.com/snakers4/silero-vad, 2021. 6. Gerganov, Georgi. Whisper.cpp. https://github.com/ggerganov/whisper. cpp, 2023.